



Data-Driven Insights for Airbnb Listings: NYC

For Data Analysis Managers and Lead Data Analyst

Objective and Agenda

Objective:

Provide insights into property pricing, neighborhood popularity, and review trends for Airbnb listings in New York City.

Agenda:

•Property Pricing and Distribution:

- Insights into the types of properties listed (entire home, private room, shared room) and their distribution.

•Neighborhood Popularity:

- Analysis of the most popular neighborhoods based on host count and review volume.

•Review Trends:

- Examination of review counts per property type and trends in customer engagement.

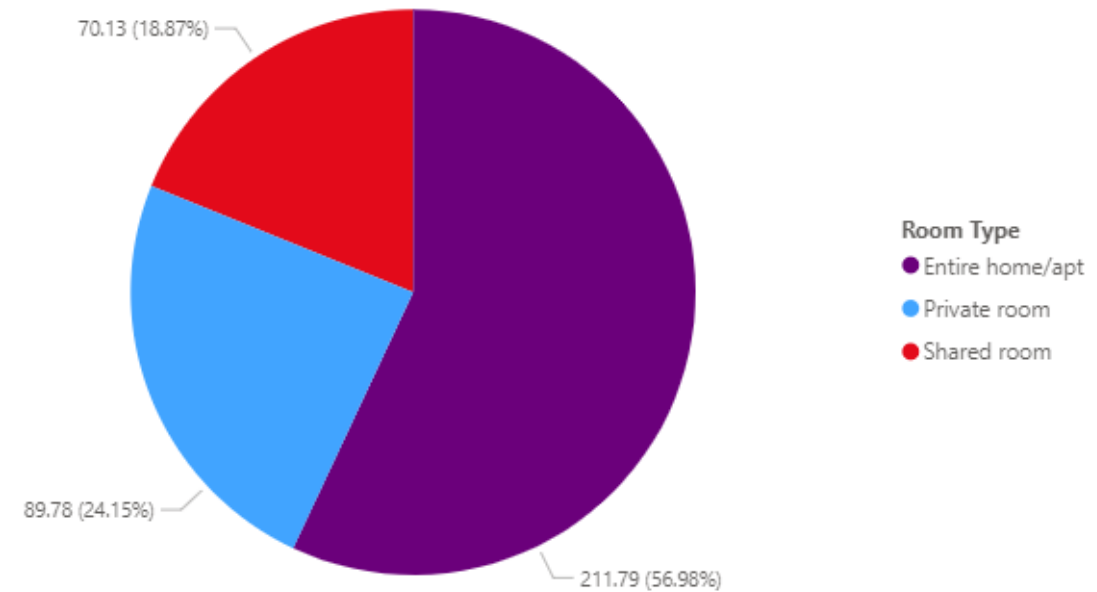
Property Pricing and Distribution

•Visualization 1:

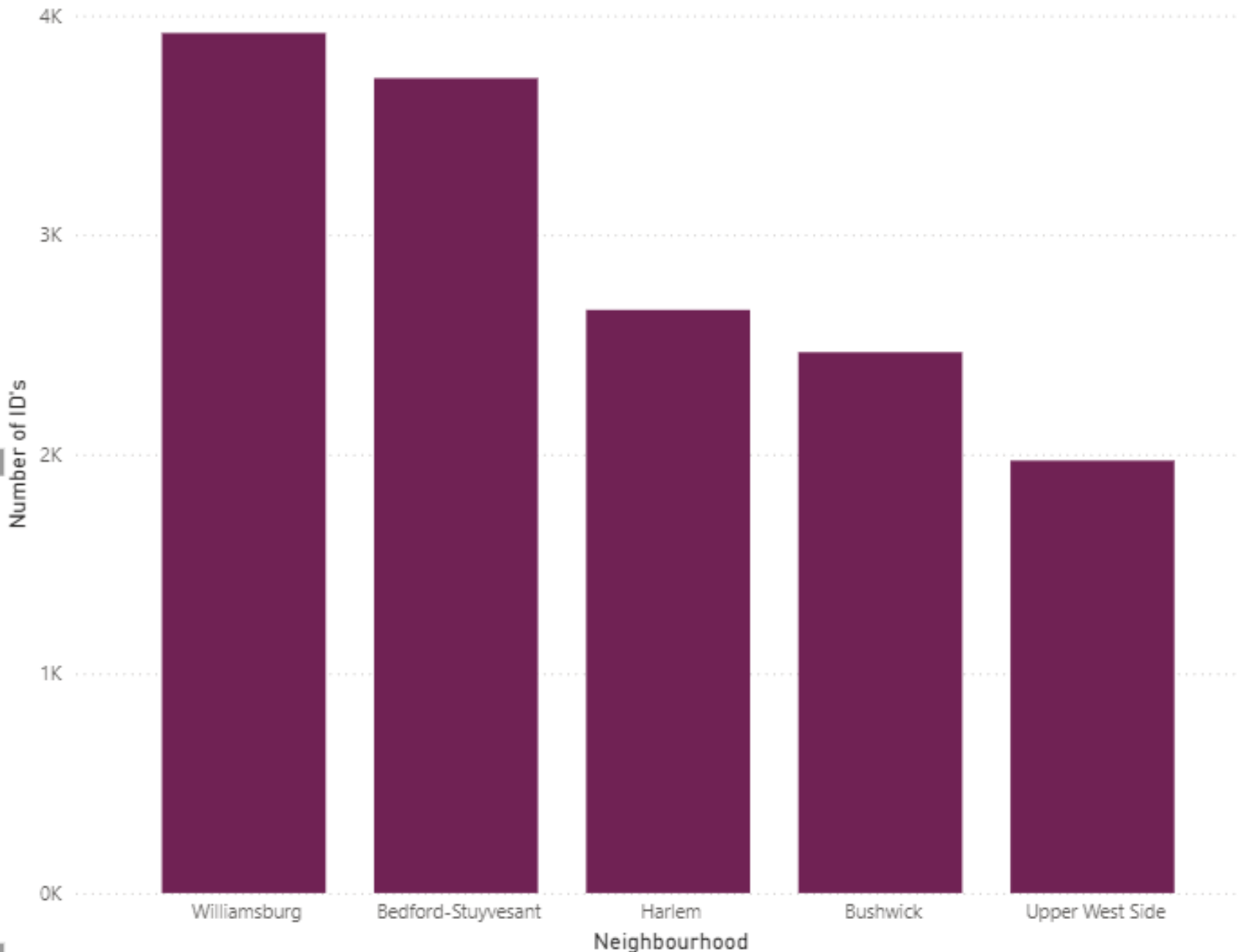
Property distribution by price (Pie Chart)

Insight: Entire home/apt makes up the majority at ~57%. Private and shared rooms are also significant.

Property Distribution by Price



Top 5 Neighbourhoods



Neighborhood Popularity

•Visualization 2

Top 5 neighborhoods based on host ID (Bar Chart)

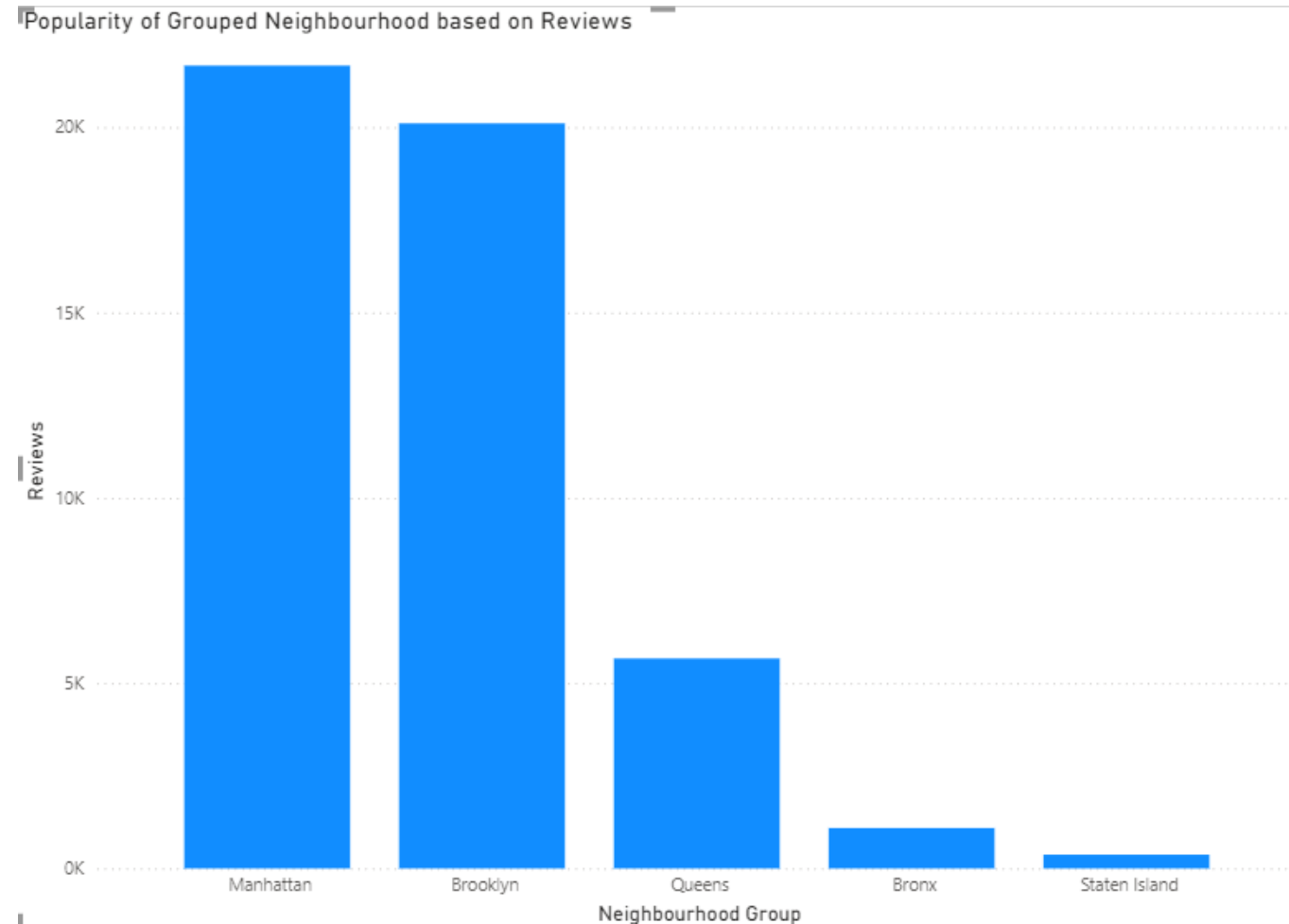
Insight: Williamsburg has the highest count of hosts, followed by Bedford and Harlem.

Grouped neighbourhood

- Visualization 3:

Popularity of grouped neighborhoods based on reviews (Bar Chart)

Insight: Manhattan and Brooklyn dominate in terms of reviews, reflecting strong demand and engagement in these areas.



Conclusion

•Key Findings:

Property Distribution: The majority of Airbnb listings are entire homes/apartments (~57%), followed by private and shared rooms. This reflects diverse customer preferences in New York City.

Neighborhood Insights: Williamsburg, Bedford, and Harlem have the highest number of hosts, while Manhattan and Brooklyn dominate in reviews. This indicates strong demand in these areas.

Review Trends: Properties with a high total review count but lower monthly reviews indicate potential fluctuations in customer engagement over time.

•Summary:

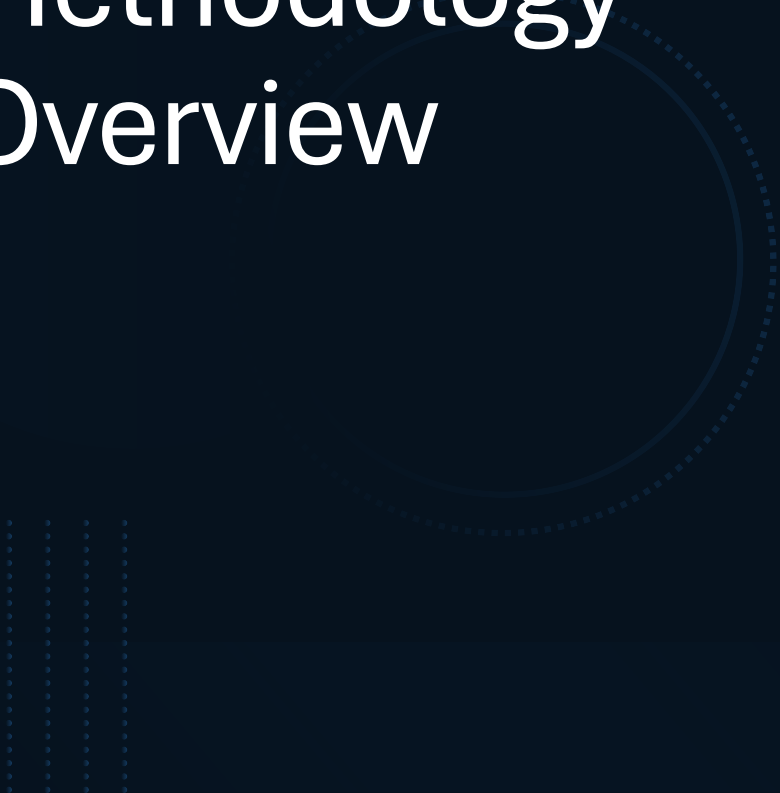
The analysis provides a clear picture of property distribution, neighborhood popularity, and review trends in New York City.

These insights can serve as a foundation for deeper exploration into customer preferences and areas of growth.

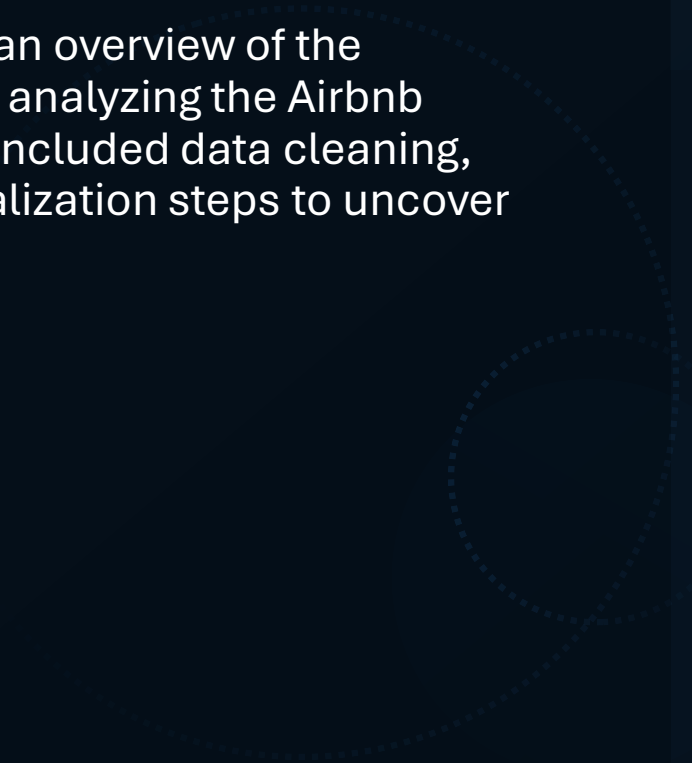
•Next Steps:

- Further investigation into customer engagement and review patterns.
- Exploration of specific neighborhoods to understand demand drivers.

Appendix: Methodology Overview

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This section provides an overview of the methodology used for analyzing the Airbnb dataset. The process included data cleaning, preparation, and visualization steps to uncover actionable insights.

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Data Cleaning

- **Data Cleaning:** We replaced missing or erroneous values (e.g., #NAME?) with UNKNOWN and removed rows with critical null values to ensure the integrity of our dataset.

Data Preparation

- The data was grouped by neighborhood, and existing metrics like "reviews per month" were used for trend analysis. Outliers in price and reviews were checked.

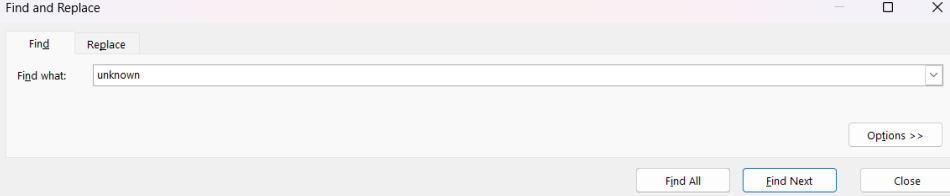
Visualization Creation

Developed three key visualizations:

1. **Property Distribution by Price:** Identified preferences for entire homes, private rooms, and shared rooms.
2. **Top Neighborhoods by Hosts:** Highlighted the most active areas based on host counts.
3. **Neighborhood Popularity by Reviews:** Ranked neighborhoods based on total reviews.

Summary of Key Steps

- Summarized key data cleaning and visualization processes.
- Provided snapshots of cleaned data (e.g., UNKNOWN replacements) and transformation results.
- Reinforced findings with clear, actionable insights.



The image shows the 'Find and Replace' dialog box in Microsoft Excel. The 'Find' tab is selected. The 'Find what' field contains the text 'unknown'. The 'Options >>' button is visible. Below the dialog box, a table displays the search results across multiple sheets.

Book	Sheet	Name	Cell	Value	Formula
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AB_NYC_2019.xlsx	AB_NYC_2019		\$B\$10324	UNKNOWN	
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AB_NYC_2019.xlsx	AB_NYC_2019		\$D\$38878	UNKNOWN	

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Detailed Methodology Document

- For full details of the methodology, including step-by-step processes and code references, please refer to the standalone methodology document provided separately.





Thank you

By Mohammed Asad